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Hypoxic adaptations of hemoglobin in Tibetan chick embryo: High oxygen-affinity mutation and selective expression

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Abstract

Tibetan chicks (*Gallus gallus*) survived with high hatchability (35.0%) and Recessive White Feather broilers (RWF) from low elevations survived rarely and with a low hatchability (3.0%) after simulated incubation under hypoxia of 13% O_2. The functional mutation of Met-32D (B13)-Leu of α^D globin chain was related with hypoxia based on allele distribution, homology model building and oxygen affinity assay. Whole embryos on days 3–8 and whole blood on days 9–18 were collected to investigate the stage expression profiles of all seven globins and HIF-1α by real-time PCR. Under hypoxia (12.0% O_2) on days 3–8, HbE was overexpressed, HbA was expressed earlier and HbP expression was restricted, which completely overturned the expression profile under normoxia. The amount of hemoglobin expression in Tibetan chicks was remarkablely higher than that of RWF. HIF-1α expression peaked early in both breeds, with. In conclusion, the special hypoxic expression profile on days 3–8 certainly is a common molecular mechanism of hypoxia tolerance in surviving Tibetan chick and RWF embryos; the mutation Met-32D(B13)-Leu and increasing hemoglobins are important mechanisms of hypoxia adaptation in Tibetan chick embryos, and we suggest that HIF-1α could be responsible for the hypoxic expression profile.
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Keywords: Tibetan chick; Embryo; Hypoxia; Hemoglobin; Hatchability; HIF-1α

1. Introduction

Tibetan chick, as an indigenous chicken (*Gallus gallus*) bred at high altitudes, is well known for its high hatchability under conditions of reduced oxygen supply. The hatchability of low land breeds is approximately zero under hypoxia at simulated altitudes of above 4000 m (Visschedijk, 1980, 1985), while the natural hatchability of Tibetan chick can reach 70–80%. We devised an incubator that can automatically control oxygen concentration, to investigate the difference of hypoxia adaptation between Tibetan chicks and Recessive White Feather broilers (RWF).

The high oxygen affinity of mutated hemoglobin is an important molecular mechanism of hypoxia adaptation in high altitude species. Bar-headed geese (*Anser indicus*) live at an altitude of 4000–6000 m and migrate annually across the summit of Mt. Everest to the Bengal Gulf. The mutation Pro-119α(H2)-Ala, responsible for the high oxygen affinity of the bar-headed goose Hb, leaves a two-carbon gap and relaxes the tension in T (tense) structure because Pro is mutated to smaller Ala (Perutz, 1983; Weber et al., 1993; Zhang et al., 1996; Liang et al., 2001). Other birds such as Andean goose (*Chloephaga melanoptera*), tufted duck (*Aythya fuligula*), white-headed vulture (*Trigonoceps occipitalis*) and black vulture (*Aegypius monachus*) have similar hemoglobin molecular mechanisms (Jessen et al., 1991; Abbasi and Lutfullah, 2002; Lutfullah et al., 2005; Hiebl et al., 1989). However, there has been no description on oxygen affinity of Tibetan chick hemoglobins, thus we have sequenced all exons of hemoglobins of Tibetan chick and RWF, determined blood-oxygen affinity and predicted the function of hemoglobin mutation with reference to hypoxia adaptation.

Seven types of hemoglobins exist in chicks: HbA, HbD, HbP, HbM, HbE, HbL and HbH, which are composed of $\alpha_2^A\beta_2^A$, $\alpha_2^D\beta_2^A$, $\pi_2\rho_2$, $\alpha_2^D\varepsilon_2$, $\alpha_2^A\varepsilon_2$, $\alpha_2^D\beta_2^H$ and $\alpha_2^A\beta_2^H$, respectively. The

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primitive erythroid cells produce five globins ρ, ε, π, α^D and α^A, and the definitive erythroid cells, replacing the primitive erythroid cells at about day 5, mainly express adult globins β^A, β^H, α^A and α^D. The embryonic ρ globin is most actively expressed at the sixth day and the fetal β^H globin is expressed from day 17 to 22 (Brown and Ingram, 1974; Chapman et al., 1981, 1982a,b; Landes et al., 1982; Cirotto et al., 1987). Obviously, the types and amounts of chick embryo hemoglobins are gradually changed with increasing development age. However, it is unclear if the individual types of hemoglobins adaptatively change in chick embryo under hypoxic condition. Therefore, we have analyzed and determined mRNA hypoxic expression profile of all seven globins and HIF-1α (Hypoxia-inducible transcription factor-1α) (Takahashi et al., 2001; Etchevers, 2003) in Tibetan chick and RWF using real-time PCR to further reveal molecular mechanisms of hypoxia adaptation in Tibetan chick.

2. Materials and methods

2.1. Sources of samples and simulated incubation under hypoxia

Two groups of Tibetan chicks in incubation experiments were raised in the chick farm of China Agricultural University, one of which was from different altitudes in Tibet and another from a high altitude of 2600 m in Tibet. The fertilized eggs of RWF were gathered from the chick farm of China Agricultural University. The blood or tissue of adult individuals was randomly collected. For Tibetan chick, 28 individuals were from altitude of 3600 m in Tibet, 24 individuals from altitude of 3000 m in Yunnan province and 22 individuals were from altitude of 2600 m in Tibet. In addition, 50 individuals of Guangxisanhuang chick and 50 individuals of Beijingyou chickens were collected from the chick farm of China Agricultural University.

Hypoxia incubator was devised by refitting a small incubator to control O_2 concentration by changing N_2 concentration but keeping other incubation conditions normal with temperature of 37.8 °C, relative humidity of 70%, and turning eggs every 2 h. The oxygen concentrations of the two incubations were 13.0% and 12.0%. During hypoxia incubation the incubator was kept airtight to maintain the oxygen concentration. The death time was identified after opening the eggshell on day 21.

2.2. PCR amplification and sequencing

Based on chick α-like globin sequences (Genbank accession no. AY016020) and β-like globin sequences (Genbank accession no. L17432), 19 primer sets were designed to amplify the seven globin genes containing all 21 exons. These primer sequences are showed as follows and A, D, P, B, R, E and H represent the primers of α^A, α^D, π, β^A, ρ, ε and β^H globins, respectively.

A11, 5′GGCCAGCACAGCATATAAGG3′; A12, 5′ TAAGGCAGGGAGGGATAGGA3′.
A21, 5′CCGAGACCCTGGAAAGGTAG3′; A22, 5′ AAAAGGAGAGGATGAGCGAG3′.
A31, 5′CACCCCCTCGCTCATCCTCTCC3′; A32, 5′ CATCTCATTTGGCTGCTCGCTGTC3′.
B11, 5′ACAACCACACGCTACCCTCCAACC3′; B12, 5′ CTCTCCCACCTCCGACCCTGAAC3′.
B21, 5′CACATTGCGCATTTTGA3′; B22, 5′ CCGAGCTGCACAGGGACACA3′.
D11, 5′ACCGCCGCTCCCTGCTCTCA3′; D12, 5′ TGCCCCCTGCCCCAACC3′.
D21, 5′AGGGCAGGGCAGGGCAGCAGT3′; D22, 5′ AGGTCGAAGTGGGGGAAGTAGGTC3′.
D31, 5′CTGCGGGTCCTGGGGTCCTCA3′; D32, 5′ GCTGCAGCAATGGTGTCTTTA3′.
P11, 5′GGCCCGCTCTCCTCTTCCTC3′; P12, 5′ TTCATAGCACGGGTGTAACTCCAG3′.
P21, 5′TAATATAGGAAACGTGGTGTCAGT3′; P22, 5′ ATCCAGTCTCAGGTTAAAAATCAG3′.
P31, 5′ TAGAGGCCACAGGTCATTTAG3′; P32, 5′ AAGCCATTTATCTGTATTTCTCA3′.
R11, 5′GCCGCCTGTTAAGCTGGTCAC3′; R12, 5′ AGCGCTGCCTCCTAAAACTGC3′.
R21, 5′GTGTGCTTGTGTCCCCCGTCTC3′; R22, 5′ ACTCCCCTCCTGCTTCCCTCCTTC3′.
R31, 5′TGCTTACCCCATTGCTCCCCTTC3′; R32, 5′ CTGCCCCCACATGTCCCCTGA3′.
E11, 5′CACCCCGACCACCCCCTCCTG3′; E12, 5′ AGCACTGCCTCCTAAAACTG3′.
E21, 5′GTGTGCTTGTGTCCCCCATCC3′; E22, 5′ CAGCCTCACAATCCTTTCTCCTTC3′.
E31, 5′CCAAACCGGCACCCTAAGA3′; E32, 5′ CCCCATACATCCCCCAAAACAGAC3′.
H11, 5′TGGCCCCACTTCCTACTCATCG3′; H12, 5′ CGCCTGCTTGCATCCCTCCTC3′.
H21, 5′AGCCAGGGGAGGGGAGAAATGAAG3′; H22, 5′AGCCCAGCAGACCCCGAGGATG3′.

50 ng of total DNA of blood was used to PCR amplification in 50 μL of PCR mixture containing 10 pM each of primer, 0.2 mM dNTP, 1.5 mM $MgCl_2$ and 2U *Taq* DNA polymerase. PCR was performed under the following conditions: DNA denaturation at 95 °C for 5 min, then 35 cycles of denaturation at 94 °C for 1 min, annealing at 55–58 °C for 1 min, and extension at 72 °C for 1 min. The purified PCR products were directly sequenced using the ABI PRISM Dye terminator Cycle Sequencing Ready Reaction Kit (Applied Biosystems) and were electrophoresed on an ABI 377 PRISM DNA sequencer (Applied Biosystems).

2.3. Gene quantification

Specimen was collected from chick embryos incubated under hypoxia of 12.0% O_2. The whole embryo on days 3–8 included allantochorion and whole blood. The blood on days 9–18 was obtained by puncturing through veins of allantochorion. The blood on days 9–18 was directly digested using Trizol reagents (Invitrogen) and the frozen whole embryo was totally

Table 1
The stage mortality and hatchability under hypoxia of 12.0% and 13.0% O_2

Oxygen concentration	Breed	Stage (day)					Hatchability
		1–3	4–8	9–11	12–17	18–21	
13.0%	Tibetan chick	4/77	1/73	2/72	17/70	26/53	35.1%
	RWF	14/234	40/220	32/180	66/148	75/82	3.0%
12.0%	Tibetan chick	3/36	1/33	2/32	12/30	14/18	11.1%
	RWF	11/171	104/160	36/56	19/20	1/1	0.0%

Number of deaths/number of sample.

grinded and partially digested. DNA was removed from total RNA using Dnase I (Ambion). Reverse transcription (RT) reaction was performed in a final volume of 20 μl using MMLV Reverse Transcriptase (Stratagene).

cDNA special primer sets were designed using ABI PRISM 7000 (Applied Biosystems). Based on chick α-like globins sequence (Genbank accession no. AY016020) and β-like globins sequence (Genbank accession no. L17432), seven primer sets were designed to quantitate the seven globin mRNAs. HIF-1α quantitation primer set was designed based on described sequences (Genbank accession nos. AB013746 and XM420327). The expression of 28S rRNA was not affected by Hypoxia (Zhong and Simons, 1999) and it was selected to be endogenous control. 28S rRNA quantitation primer set was designed based on described sequence (Genbank accession no. M59792). Primer sets are.

α^A: 5′TCCGCTGCTGACAAGAACAAC3′; 5′ GGGGGTAGGTGGTGAA CATCC3′.
α^D: 5′CTGACTGCCGAGGACAAGAAGC3′; 5′ AGGTCGAAGTGGGGGAAGTAGGTC3′.
π: 5′CCAGCTATCCTCAGACGAAAACCT3′; 5′ TTGGCCAAAGCACCTCTAATGTC3′.
β^A: 5′TTTGGGGATGCTGTGAAGAAC3′; 5′ GGGCGGCCAGGACAATGATGA3′.
β^H: 5′TTGGGGAGGCTGTAAAGAA3′; 5′ GGGAGGCCAGGACGATGAT3′.
ε: 5′TTGCACGTGGACCCCGAGAAC3′; 5′ GCAAATTGGCAGGCAGGAGTGAAA3′.
ρ: 5′TTTGGGGAAGCCGTGAAGAAC3′; 5′ CTGGCAGGTCGGGGTGAAGTC3′.
HIF-1α: 5′CAGCAGCAGTAGAAGTTGTGATAG3′; 5′ AAATGTGTAACGAATGGGGAAGAT3′.
28s rRNA: 5′GGAGCCCCGGGGAGAGTTC3′; 5′ GGATTTTCACGGGCCAGCGAGAG3′.

Gene expression was measured by quantitation of cDNA relative to a calibrator sample and real-time PCR was conducted with an initial denaturing at 50 °C for 2 min, and then involved 40 cycles of 95 °C, 15 s; 55 °C, 45 s; 72 °C, 30 s using SYBR green PCR Master Mix on Applied Biosystems 7900HT Fast Real-Time PCR System (Applied Biosystems). Gene-specific standard curves for 28S rRNA, HIF-1α and seven globin cDNAs were generated from serial 10 fold dilutions of the RT-product mixture.

2.4. The blood-oxygen affinity and BPG concentration

The whole blood of 15-day-old embryos was obtained in 5 min by puncturing veins of allantochorion and five embryos were pooled for whole blood oxygen affinity assay. The whole blood of adult chick, at four months of the age, was taken from the brachial vein. The hemoglobin affinity for oxygen was determined by the mixing technique (Edwards and Martin, 1966; Scheid and Meyer, 1978). This method required samples of blood equilibrated with nitrogen and with a hyperoxic mixture at 41 °C in a tonometer. The Po_2, Pco_2 and pH of the mixed blood sample were determined in ABL555 Blood gas analyzer (Radiometer). The Bohr factor used was −0.45 for the 15-day-old embryo blood and −0.54 for adult blood (Reeves, 1984). BPG (2,3-bisphosphoglycerate) concentration was determined with enzymatic assay kit from Sigma-Aldrich Chemical Co. BPG was related to the molar concentration of tetrameric hemoglobin rather than to erythrocytic volume to avoid possible inaccuracies in volume determination (Ingermann et al., 1983).

2.5. Prediction of proteins structure

The tertiary and the quaternary structure of hemoglobins were predicted and built by SWISS-MODEL server approach (Schwede et al., 2003) (http://swissmodel.expasy.org) using crystal structure coordinates of chick hemoglobin HbD (PDB code=1hbr). The visualization and analysis of protein structures were carried out using Swiss-Pdb Viewer (V3.7) (http://www.expasy.org/spdbv) and DS ViewerPro5.0 program of Insight (95.5) soft package (http://www.accelrys.com/viewer).

2.6. Data analysis

Chi-square test of two and three dimensions of incubation data was carried out using CATMOD process of SAS8.2 (SAS Institute Inc.). Primers were designed and sequences were edited using DNASTAR5.0 (DNASTAR Inc.).

3. Results

3.1. Hypoxia incubation

Table 1 displays the stage mortality and hatchability of Tibetan chick and RWF in the two incubation experiments under hypoxia of 13.0% and 12.0% O_2, respectively. The

Table 2
The mutation frequency of α^D gene in Tibetan chicks at different altitudes

Altitude (m)	AA	AC	CC	Allele C frequency
2600	10	2	10	50.0%
3000	0	8	16	83.4%
3600	2	1	25	91.1%
2600–4200	7	3	18	69.6%

AA denotes wild type, CC denotes mutation type, and AC denotes heterozygote.

significant interaction ($\chi^2=25.86$, $P<0.001$) between breed and stage indicated that the ability to tolerate hypoxia most significantly differed on days 4–11 of incubation in that a high amount of RWF individuals died at this stage while Tibetan chick began to die after day 12. Under hypoxia of 13.0% O_2, RWF rarely survived with hatchability of 3%, while under hypoxia of 12.0% O_2, Tibetan chick exhibited low hatchability of 11.1%.

3.2. Hemoglobin mutation

The mutation Met-32D(B13)-Leu was determined in α^D chain in Tibetan chick, but it was not found in 50 Beijingyou chicks, 50 Guangxisanhuang chicks and 7 RWFs surviving from 234 individuals incubated under hypoxia of 12.0% O_2. The frequency of mutation Met-32D(B13)-Leu is shown in Table 2. Obviously, the gene frequency of mutated α^D increased with increasing altitude. This mutation appeared to be fixed with a high frequency of 91.1% in the group at high altitudes of 3600 m, and occurred with a much lower frequency of 50.0% in the group at high altitude of 2600 m.

3.3. The structure of mutated hemoglobins

The mutation Met-32D(B13)-Leu of HbD, HbM and HbL occurs in hydrophobic heme pocket. The hydrophobicity is stronger in Met with a polar sulfur atom than in Leu, equal to 1.90 and 3.80, respectively, therefore the substitution Met-32D (B13)-Leu serves to stabilize the oxy-structure, providing more hydrophobic environment.

Table 3
The normalized mRNA amounts of globin chains in Tibetan chick and RWF

Stage (day)	Breed	α^D	π	α^A	β^A	ε	ρ
3	B	0.11	0.04	1.74	1.55	1.49	0
	Z	1.70	2.17	60.37	2.55	3.92	0
4	B	0.23	0.07	1.66	4.43	8.29	8.99
	Z	1.46	2.68	6.69	28.82	70.52	26.78
5	B	0.43	0.42	8.46	4.18	3.83	0.89
	Z	1.17	0.29	46.78	64.93	2.19	11.63
6	B	0.04	0.02	2.31	3.10	2.43	0.01
	Z	30.08	3.99	134.35	136.80	73.70	56.24
7	B	0.06	0.02	3.38	4.61	1.18	0.02
	Z	2.29	0.38	10.57	5.02	4.41	3.70
8	B	0.21	0	6.72	29.70	0.97	1.64
	Z	0.79	0.03	9.40	35.12	0.17	3.92

"0"denotes trace; "Z" and "B" denotes Tibetan chick and RWF, respectively.

3.4. The blood-oxygen affinity and concentration of BPG

The blood-oxygen affinity (P50, Po_2 required to achieve a saturation of 50% at pH 7.4 and 41°C), for 15-day-old embryos ($n=15$) of Tibetan chick and RWF, were 37.7 mmHg and 44.7 mmHg, respectively, and for adult individuals ($n=15$), 51.8 mmHg and 58.7 mmHg, respectively. Under normoxia, BPG contents of 15-day-old embryos of Tibetan chick ($n=8$) and RWF ($n=10$) were 1.02 and 0.87 (mol BPG/mol hemoglobin tetramer), respectively, and did not differ significantly ($P>0.05$). Under hypoxia, BPG (2, 3 diphosphoglycerate) contents of 15-day-old embryos of Tibetan chick ($n=8$) and RWF ($n=8$) were 1.10 and 1.09 (mol BPG/mol hemoglobin tetramer), respectively, and also did not differ significantly ($P>0.05$).

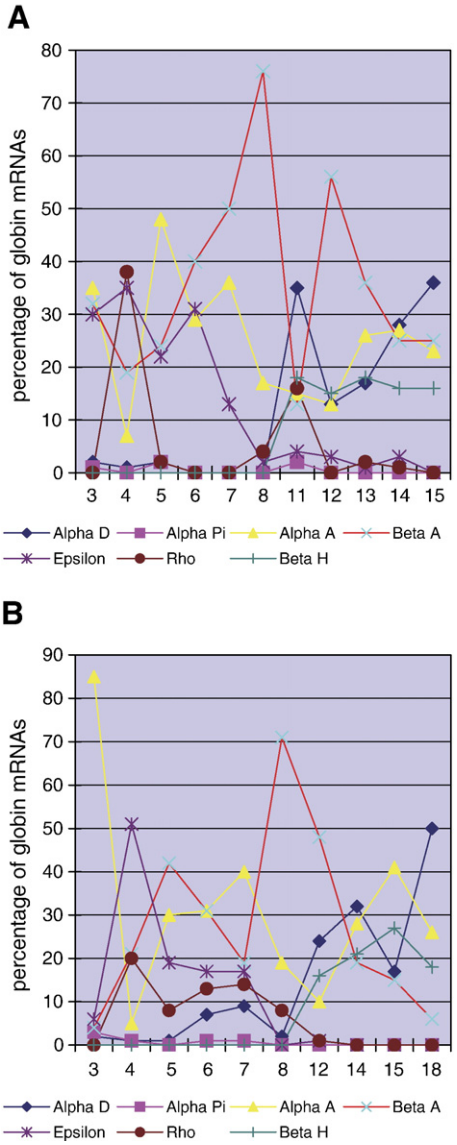


Fig. 1. The stage (day) mRNA levels of seven globins. *Globin* gene expressions were presented as percentage of goobin mRNA in the whole embryo on days 3–8 and in the whole blood on days 9–18. (A) The stage (day) mRNA levels of seven globins in RWF. (B) The stage (day) mRNA levels of seven globins in Tibetan chick.

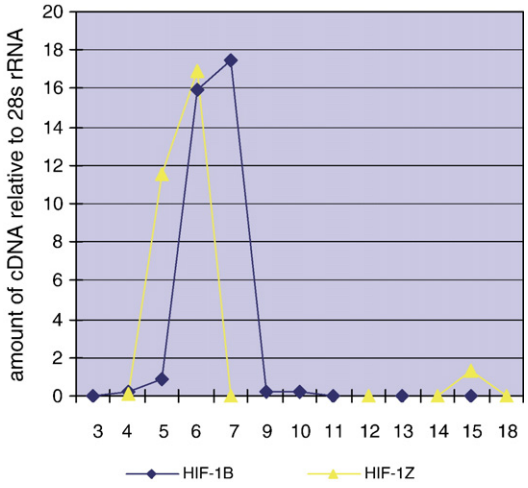


Fig. 2. The stage (day) mRNA levels of *HIF-1α* in Tibetan chick (Z) and RWF (B). *HIF-1α* gene expressions were presented as ratio of mRNA relative to *28s rRNA* in the whole embryo on days 3–8 and in the whole blood on days 9–18.

3.5. Globin mRNA levels

The normalized expression amount of six globins in the embryos on days 3–8 under hypoxia of 12.0% O_2 is listed in Table 3. The expression amount of all major globins on days 3–8 in Tibetan chick was several, even up to 50 folds, higher than in RWF.

The amount of globins expressed in the same stage was presented as copies of cDNA without normalization because their mRNA samples were same. The percentage of each globin mRNA in Tibetan chick and RWF embryos is shown in Fig. 1. In 3–8-day-old embryos, major adult globins α^A and β^A over-expressed and became major types, major embryonic globins π and ρ became minor types, and minor embryonic globin ε became a major type. In embryonic blood of 9–18 day, globin β^H expressed with a great deal, the expression level of globin α^D increased remarkably, globins ε, π and ρ still expressed with trace amount, and globins α^A and β^A still expressed with a great deal.

3.6. HIF-1α mRNA level

The normalized mRNA amounts of HIF-1α in Tibetan chick and RWF embryos are shown in Fig. 2. HIF-1α mRNA was expressed throughout hatch period under hypoxia. The expression peaks of Tibetan chick and RWF occurred on day 5, 6 and on day 6, 7, respectively, and the former is one day earlier than the latter.

3.7. The structure predicting of functionally variant sites of hemoglobins

According to BPG contact sites and subunit contact sites of human hemoglobin (Perutz, 1970a,b, 1983; Shih et al., 1993; Rollema and Bauer, 1979; Chapman et al., 1980; Imai et al., 1989; Arnone, 1972; Kleinschmidt and Sgouros, 1987; Perutz et al., 1987), we have aligned and analyzed the corresponding variant sites of seven hemoglobins of chick. For subunit contact sites (see Table 4), strong van der Waals interaction occurs at $\alpha_1^A\varepsilon_1$ contact sites α^A34Thr with ε128Phe and α^A35Thr with ε128Phe in HbE, the same interaction occurs in HbM and weak interaction also occurs in HbA, HbH, HbL and HbD. HbP loses van der Waals interaction at $\pi_1\rho_1$ subunit contact sites π34Ala with ρ124Pro, π35Ser with ρ128Ala, ρ127Gln and ρ131Gln and $\pi_1$119Pro with $\rho_1$55Ile arising from the distances more than 4 Å between these contact sites. HbL and HbH also lose van der Waal interaction at $\pi_1$119Pro with $\rho_1$55Ile. In addition, only HbP loses hydrogen bond at site $\pi_1$103His with $\rho_1$131Gln. The electron density map of amino acid residues around BPG contact sites in HbP reveals that substitutions of Arg-ρ143-Tyr and Arg-ρ135-Ser reduce positive charge around BPG (Fig. 3A). In contrast, the similar substitution Arg-ε135-Asn in HbM and HbE has a slight change of charge around BPG. It is suggested that the key BPG binding site is not ε135 but ρ143 (Fig. 3C). HbA, HbH and HbL have the same amino acid as HbD at both sites (Fig. 3B).

4. Discussion

4.1. The functional mutation Met-32D(B13)-Leu and hypoxia

The mutation Met-32D(B13)-Leu was not found in the three indigenous chick breeds, but the frequency of this mutation increased with increased altitude in Tibetan chick. We also performed incubation experiment in which two Tibetan chick groups with different mutation frequencies were incubated simultaneously under hypoxia of 12.0% O_2. The results showed that the hatchability was 12.1% when the mutation frequency was 70% and the hatchability was 4.0% when the mutation frequency was 50%. Obviously, the hatchability is positively related with mutation frequency in Tibetan chick. In conclusion, the mutation Met-32D(B13)-Leu is functionally responsible for hypoxia adaptation.

Homology modeling of the mutated HbD using described crystal structure (Knapp et al., 1999) revealed that this mutation, occurring in hydrophobic heme pocket (B13), leads to more hydrophobic and stable environment and increases the oxygen affinity. The blood-oxygen affinity of 15-day-old embryos and adult individuals was higher in Tibetan than in RWF. Compared with the P50 (29 mmHg) of adult Andean chick (Velarde et al., 1997), those of adult Tibetan chick and RWF were higher with 51.8 and 58.7 mmHg, respectively, consistent with other

Table 4
The altered van der Waal distances (A°) at αβ contact points

α_1	β_1	$\alpha^A\beta^A$	$\alpha^D\beta^A$	$\alpha^A\varepsilon$	$\alpha^A\beta^H$	$\alpha^D\beta^H$	$\alpha^D\varepsilon$	πρ
34	124	3.56	3.61	3.57	3.57	3.58	3.57	4.22
	128	3.84	3.83	2.79	3.87	3.86	2.80	3.84
35	127	3.72	3.47	3.72	3.72	3.72	3.71	4.18
	128	3.56	3.50	2.95	3.53	3.53	2.96	4.16
	131	3.50	3.59	3.49	3.50	3.50	3.49	4.90
103	131	H	H	H	H	H	H	3.19
119	55	3.98	3.88	3.26	4.68	4.67	3.32	4.68

"H" denotes hydrogen bond.

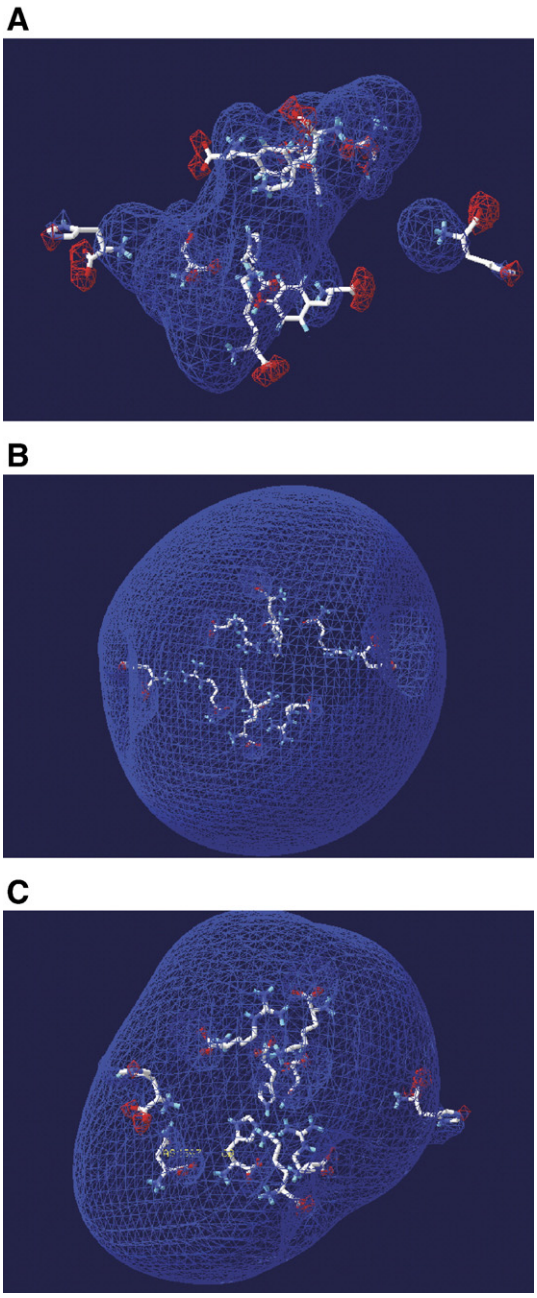


Fig. 3. Electronic density at BPG contact sites. The model was genetated using the Swiss model server and viewed in Swiss PDBViewer. (A) The reduced electronic density at BPG contact sites in HbP, including uncharged ρ143Tyr and ρ135Ser. (B) The strong electronic density at BPG contact sites in HbD, including positively charged β^{A}143Arg and β^{A}135Arg. (C) The slightly reduced electronic density at BPG contact sites in HbE, including unique site ε135Asn.

literature (Reeves, 1984). It is possible that Tibetan chick and Andean chick have different hemoglobin mechanisms although they both reside in high altitude areas. The percentage of α^{D} globin was equivalent between the two breeds (Fig. 1) in 15-day-old embryos and the estimated quantity of HbD and HbL of Tibetan and RWF amounted to approximately 50% of their total hemoglobins in this stage (Fig. 4). Structure prediction also showed that the hemoglobins expressed during intermediate and later stages have equivalent ability to bind BPG. With respect to intracellular concentration, the primary organic phosphate in early embryonic development is ATP and later, BPG with a weaker allosteric effect occurring in intermediate and later embryos with a more hypoxic condition (Bartlett and Borgese, 1976; Baumann et al., 1982, 1983), so we determined BPG concentration of day 15 embryos and the result showed that BPG concentration of these two breeds did not differ significantly under normoxia and hypoxia, respectively. Therefore the difference of blood-oxygen affinity of day 15 embryos should arise from the mutation Met-32D(B13)-Leu rather than concentration or ability to bind with BPG. Likewise, the higher blood-oxygen affinity of adult Tibetan chick, containing HbD approximately 25% of total hemoglobins (Brown and Ingram, 1974; Romanoff, 1960; Lewis et al., 1991), should also result from the mutation Met-32D(B13)-Leu. In days 3–8 embryo, the proportion of α^{D} globin (Fig. 1) in total hemoglobins of both breeds was too low to affect the affinity for whole embryonic

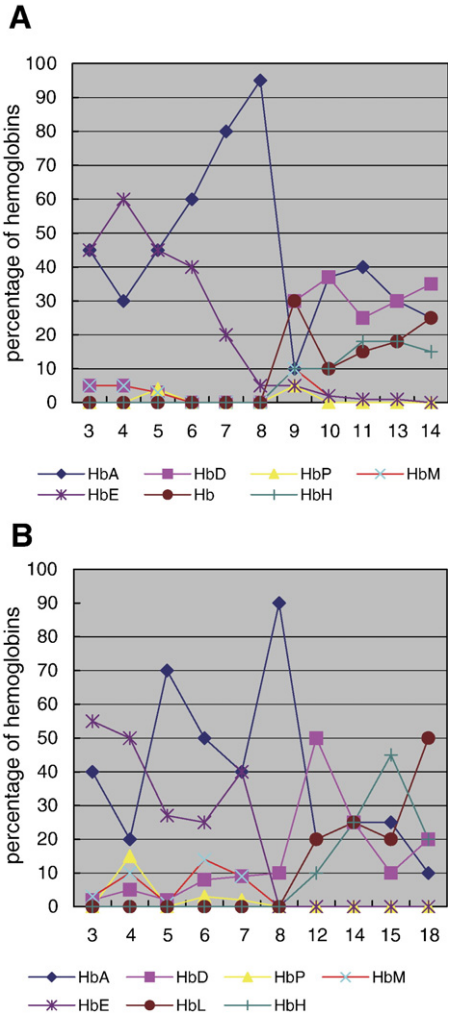


Fig. 4. The stage (day) expression profile of seven hemoglobins. Hemoglobin amounts were evaluated according to the globin subunit with the lowest expression amount and were presented as the percentage of total hemoglobin in the whole embryo on days 3–8 and in the whole blood on days 9–18. (A) The evaluated stage (day) expression profile of seven hemoglobins in RWF. (B) The evaluated stage (day) expression profile of seven hemoglobins in Tibetan chick.

blood. Consequently, the mutation Met-32D(B13)-Leu of Tibetan chick do not exert hypoxia adaptation advantage during days 3–8 of embryo but play an important role during days 9–21 of embryo and in adult individual.

4.2. Hemoglobin expression profile induced by hypoxia

Specific waves of transcription of β-like globin gene of chick genome, appearing during the interval between 5 and 12 days of incubation, had been determined by transcription *in vitro* and southern blots analysis (Landes et al., 1982). The results revealed that β^A globin mRNA began to express at a low but significant level in 5-day embryo and ρ and ε globin genes transcribed most actively in 6-day embryo. The general constitution of hemoglobins in chick embryo are as follows: a single switch event in which the primitive cells are replaced in the circulation by definitive lineage cells occurred at about day 5; the proportion of HbP: HbE: HbM is 7:2:1 in embryonic hemoglobins of days 1–5 (Bruns and Ingram, 1973); the proportion of HbA: HbD is 3:1 in adult hemoglobin of days 9–21 (Brown and Ingram, 1974; Romanoff, 1960); HbH and HbL are minor fetal hemoglobins on days 7–22 (Cirotto et al., 1987). Globin switching comes from the change between primitive and definitive cells rather than regulating transcription in the same cell linage (Dieterlen et al., 1976, Dieterlen and Martin, 1981; Maeno et al., 1985a,b; Turpen et al., 1981; Nikinmaa, 1990). Under hypoxia of 13.5% O_2, the switching between embryonic and adult globins happened earlier due to prematurity of the definitive lineage (Baumann et al., 1983). In the present studies, under hypoxia of 12.0% O_2, globin β^A expressed earlier on day 3 due to prematurity of the definitive lineage, globins ε and α^A overexpressed abnormally and globins π and ρ were excessively restrained in 3–8 day embryo. Additionally, the embryonic globins ε, π and ρ still expressed slightly on days 9–18, but their mRNAs possibly were not translated effectively (Brown and Ingram, 1974; Romanoff, 1960; Cirotto et al., 1987; Chapman and Tobin, 1979).

The expressions of both α- and β-like globins were unequal and the percentages of hemoglobins were evaluated based on globin subunit with the lowest expression amount. The evaluated stage expression profile of seven hemoglobins is shown in Fig. 4. In days 3–8 embryos of the two breeds, the adult major hemoglobin HbA is expressed earlier and became major, the embryonic minor hemoglobin HbE overexpressed and became major, the embryonic major hemoglobin HbP became minor and the embryonic minor hemoglobins HbD and HbM still maintained slight expression. The expression amount in Tibetan chick was much more than in RWF, approximately 3–6 folds at days 3, 4, 5 and 7, up to 50 folds at day 6 and equivalent at day 8. In days 9–18 blood of two breeds, the proportions of HbA, HbD, HbH and HbL were approximate serving as major hemoglobins and HbM and HbE still expressed with trace amount.

HIF-1α is stabilized in hypoxia, translocates into the nucleus and dimerizes with HIF-β, which is expressed independently of the prevailing oxygen pressure, to form HIF-1 and binds with the hypoxia response element (HERE) to express into the various genes (e.g. EP0-erythropoiesis; VEGF-angiogenesis, NOS/HO-1-vasodilation; glycolytic enzymes-anaerobic metabolism; TH-catecholamine synthesis) (Wang et al., 1995; Bunn and Poyton, 1996; Semenza, 2000). Blind subterranean mole rats (Spalax, Spalacidae) evolved adaptive strategies with hypoxia and they had significantly higher, quicker and longer responses to different O_2 levels than rattus based on gene expression of HIF-1α using real-time PCR (Shams et al., 2004). In the present studies, under hypoxia of 12% O_2, HIF-1α expressed throughout incubation and reached maximal expression at days 5–7 in both breeds. Moreover, the expression amounts at expression peak were approximately equivalent between the two breeds. Based on the time the expression peak occurs, HIF-1α gene should be concerned with the earlier switching of definitive erythroid cells and the on-off of globin genes under hypoxia, but these regulation mechanisms remain to be further solved.

4.3. The function, expression of hemoglobins in embryo and hypoxia adaptation

The fetal hemoglobin HbF of human exhibits high affinity for oxygen with a substitution His–γ143–Ser that attenuates the ability to bind BPG and so the fetal blood can obtain more oxygen from maternal blood (Bunn et al., 1977). Unexpectedly, a similar substitution Arg–ρ143–Tyr occurs in chick HbP with an uncharged Tyr residue substituting positively charged Arg residue. The electron density map around BPG contact sites showed that the substitution Arg–ρ143–Tyr reduces positive charges around BPG contact sites (Fig. 3). As a result, HbP has lower affinity for BPG mainly due to the substitution Arg–ρ143–Tyr. Additionally, HbP loses van der Waals interactions at five contact sites and a hydrogen bond at the contact site $\pi_1$103His with $\rho_1$131Gln compared with other hemoglobins. More importantly, the mutation Met-32D(B13)-Leu of α^D subunit also occurs in HbP. Undoubtedly, the substitution Met-32π(B13)-Leu and the loss of these intersubunit contacts greatly increase intrinsic oxygen affinity of HbP.

Chick embryo has relative affluent oxygen on days 4–8 compared with the later stages (Freeman and Misson, 1970; Tazawa and Mochizuki, 1977; Meuer and Baumann, 1988) and ATP is the primary erythrocytic organic phosphate early in chick embryo development (Bartlett and Borgese, 1976; Baumann et al., 1982, 1983). In this stage, HbP exerts strong ability to bind oxygen with high intrinsic oxygen affinity and low BPG affinity while HbM and HbE exert their strong ability to discharge oxygen with low oxygen affinity and high BPG affinity, so the two adaptive mechanisms concur in days 4–8 embryo. After day 8, the surrounding become more and more hypoxic, HbA, HbD, HbL and HbH serve as major hemoglobins with intermediate oxygen affinity and the blood-oxygen affinity increases with declined ATP concentration (Freeman and Misson, 1970; Tazawa and Mochizuki, 1977). However, our mRNA quantitation results showed that, in days 3–8 embryo under hypoxia of 12.0% O_2, HbE overexpressed, HbA was expressed earlier and HbP expression was restricted, which

completely overturned the expression profile under normoxia. The reasonable interpretation is as follows: 12.0% O_2 is mortal to the development of RWF and Tibetan chick embryo with no surviving individuals and hatchability of 11.1%, respectively; in such excessively hypoxic condition, HbP could not discharge oxygen efficiently in the tissues due to too high affinity for oxygen, with high intrinsic oxygen affinity and low BPG affinity, so its expression is entirely restrained; on the contrary, HbE overexpresses possibly because of the low oxygen affinity; HbA expresses earlier with a great deal aiming to increase in the total hemoglobin concentration. Accordingly, the special hypoxic expression profile on days 3–8 certainly is the common important molecular mechanism of hypoxia adaptation in survived Tibetan chick and RWF embryos.

High altitude vertebrates scarcely have erythrocytosis but high altitude bird embryos exhibit erythrocytosis (Tazawa, 1986; Carey et al., 1993; Carey and Martin, 1997). Our quantitation result showed that the expression amounts of major hemoglobins in Tibetan chick, under hypoxia of 12.0% O_2, were several, even fifty at day 6, folds higher than those in RWF on days 3–8. In extreme hypoxia, the adaptive mechanism of special hypoxic expression could not satisfy the oxygen required for chick embryonic development. Thus the much higher hemoglobin concentration is an important molecular mechanism of remarkably higher hatchability of Tibetan chick.

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